



BITS Pilani

Pilani Campus

Department of Computer Science & Information Systems

Topic -

Author(s): ____

Course Name:

Mentor:

Problem Statement

Objective: To design and develop a privacy-preserving publish-subscribe-based decentralized framework for Mobile Crowdsensing

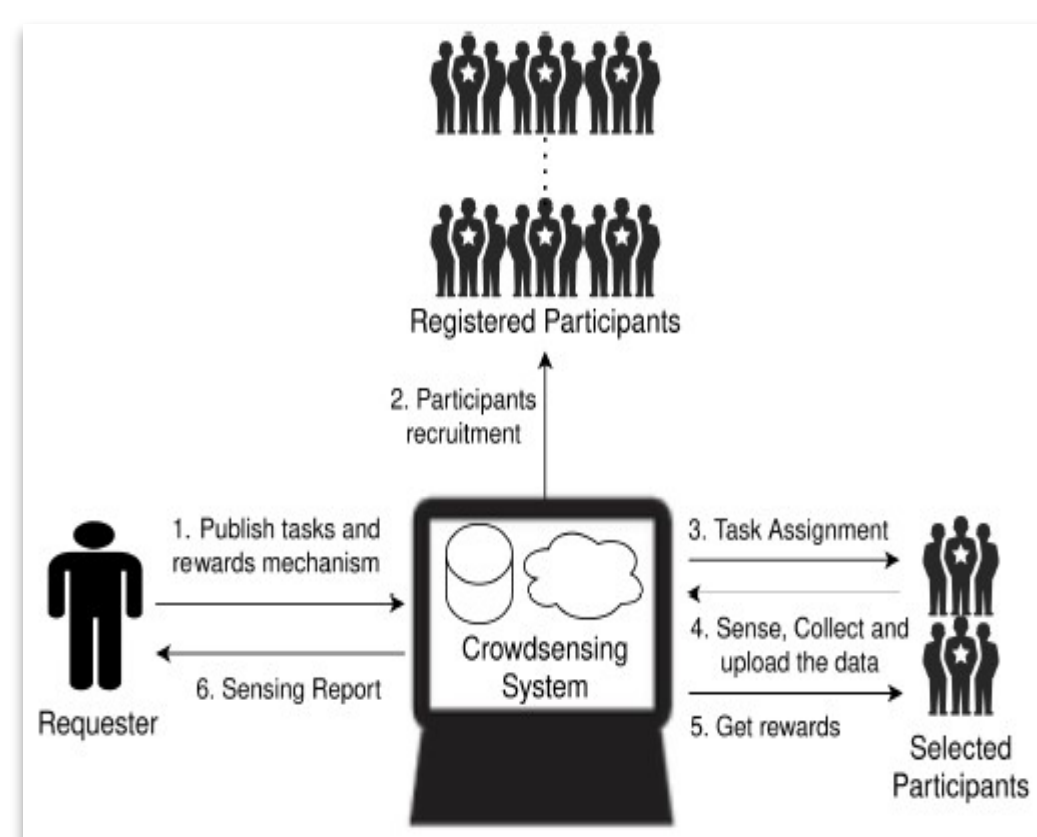


Fig 1: Overview of MCS system

Proposed privacy-preserving publish-subscribe-based decentralized framework named "Pub-SubMCS" for an MCS system

Issues with Centralized MCS systems: A single point of failure and trust issues due to a lack of transparency.

Blockchain Features: Security, Immutability, Transparency, Identity privacy, and Trust.

Proposed Framework

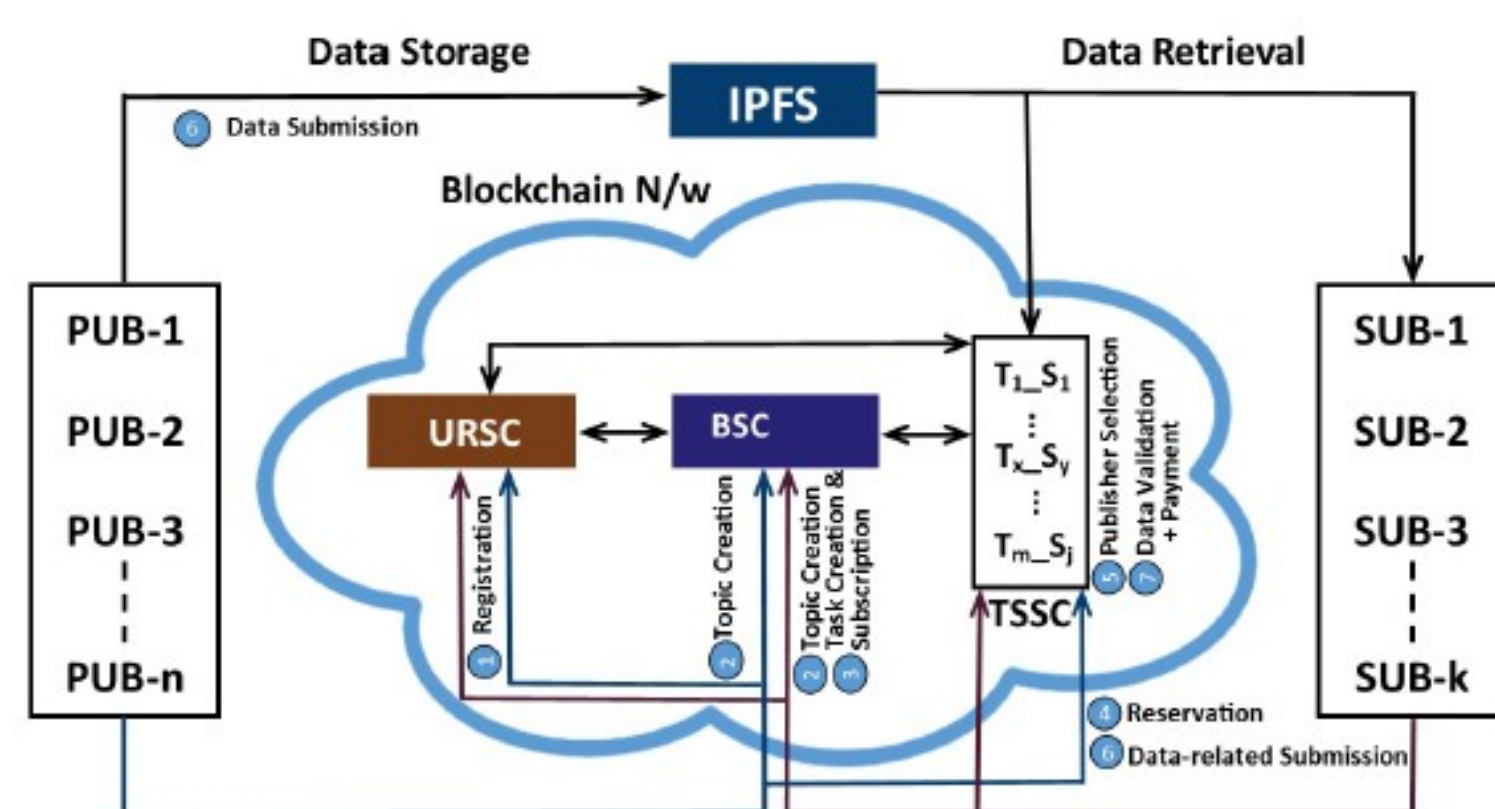


Figure 2: Decentralized Pub-Sub-based framework

System phases: User Registration, Topic Creation, Task Creation & Subscription, Publisher Reservation, Publisher Selection, Data Submission, Data Validation & Payment Distribution.

Smart contracts: URSC, BSC, and TSSC.

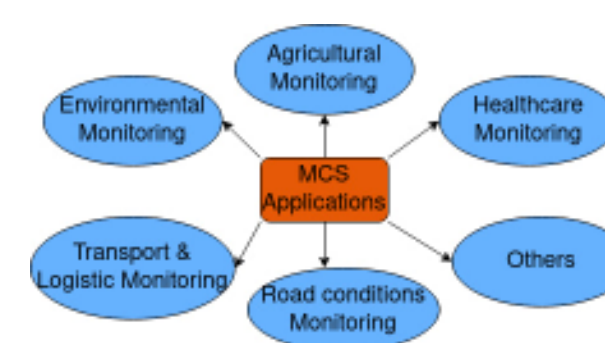


Fig 3: MCS Applications

Results

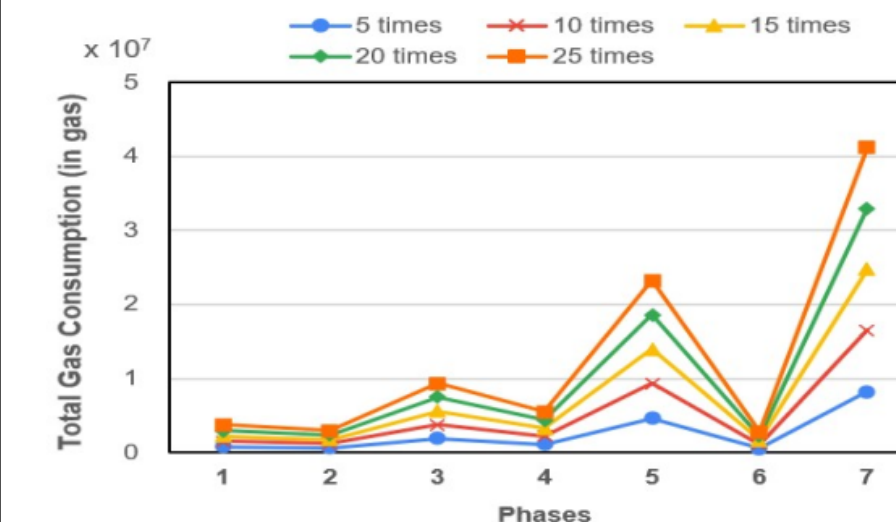


Fig 4: Total gas consumption of each phase with multiple invocations



Fig 5: Effects of number of subscribers under a task on Gas

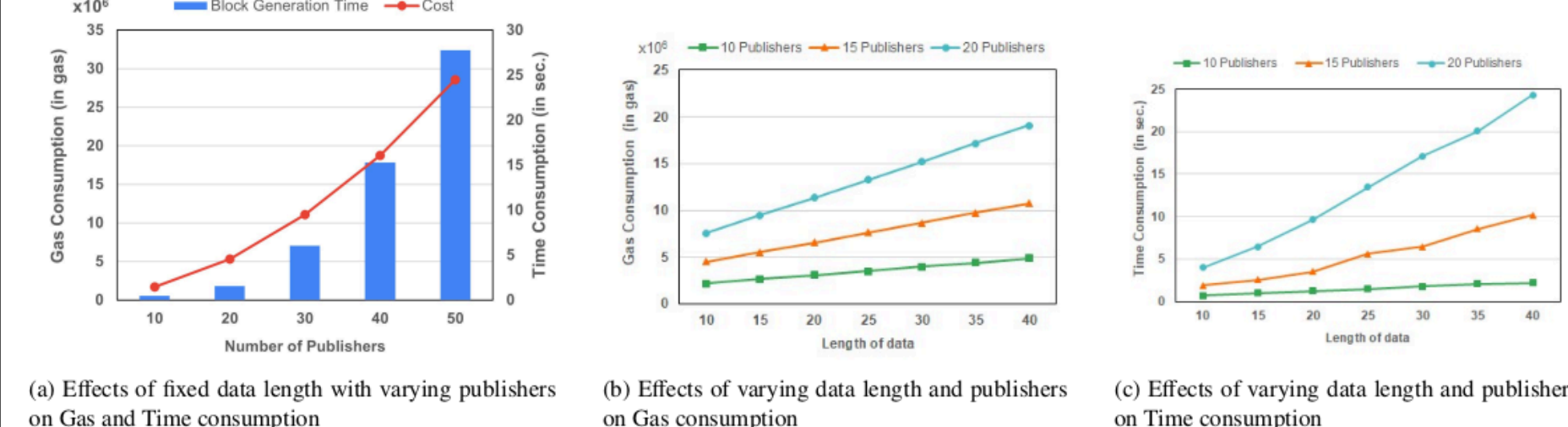


Fig 6: Effects of parameters on data validation phase

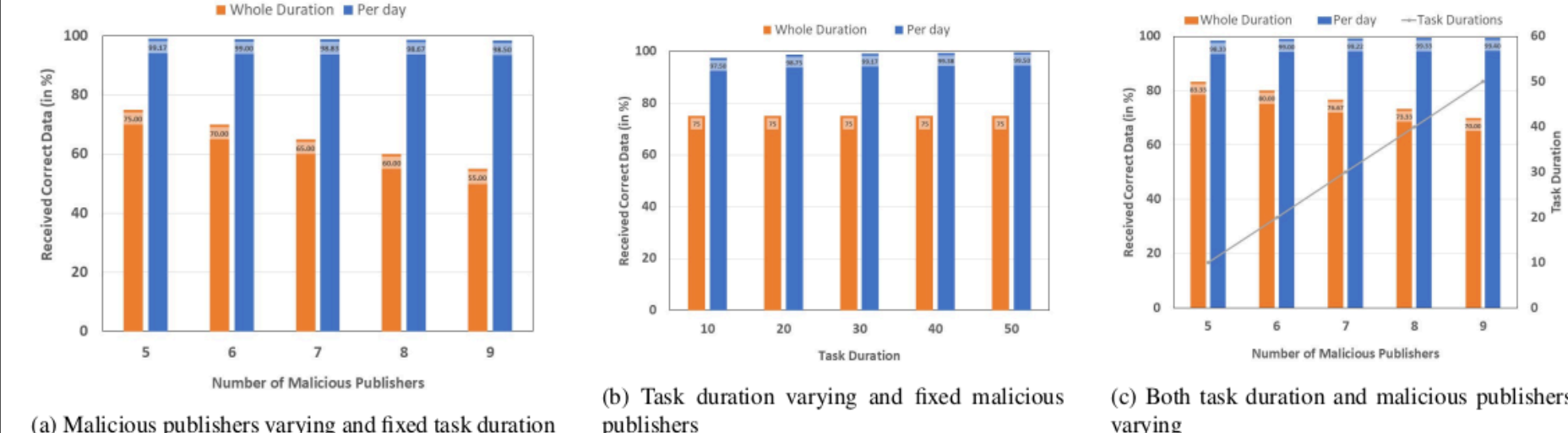


Fig 7: Effects of submitting and validating data for whole task duration vs. once per day on the curse of sensing issue

Research Gaps

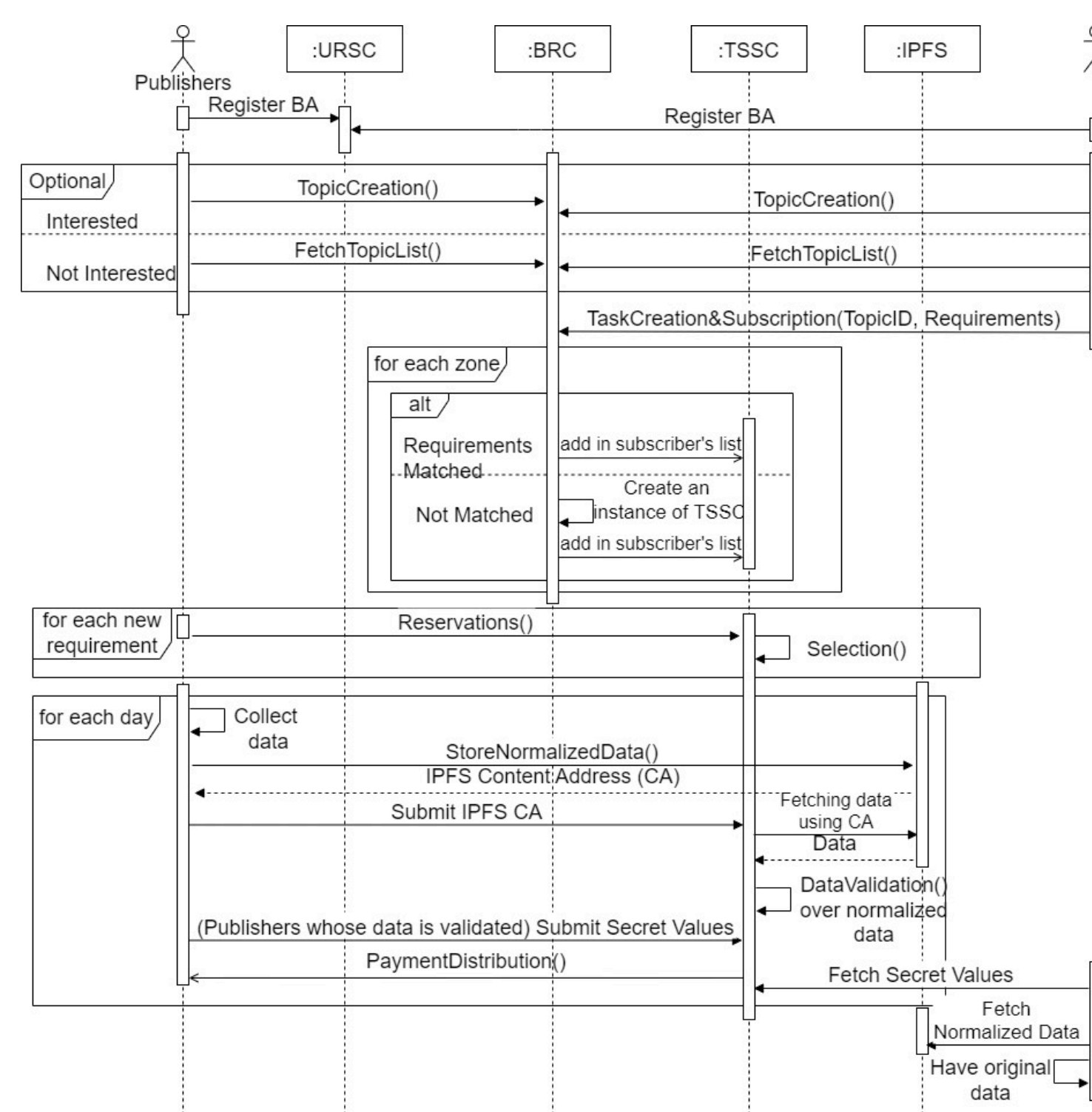
Issues with existing blockchain-based system [1-6]:

- 1) Data-sharing service
- 2) Historical Data Access
- 3) Curse of sensing issues
- 4) Privacy-preserving lightweight on-chain data validation

Contributions

- 1) Pub-SubMCS incorporates a publish-subscribe service model that allows sharing of data requests and reduces the wastage of publisher's and blockchain node's computing resources.
- 2) Pub-SubMCS incorporates a smart contract-based solution to handle the curse of sensing issues, which imposes restrictions on publishers to publish data and detects and penalizes them early if involved in malicious activities.
- 3) Pub-SubMCS incorporates a privacy-preserving on-chain data validation method by executing a similarity matching algorithm over transformed data to ensure the trust promised by a decentralized MCS system.

Activity Sequence Diagram



References

- [1] Chatzopoulos, D., Gujar, S., Faltings, B., & Hui, P. (2018, October). Privacy preserving and cost optimal mobile crowdsensing using smart contracts on blockchain. In 2018 IEEE 15th International Conference on Mobile Ad Hoc and Sensor Systems (MASS) (pp. 442-450). IEEE.
- [2] Li, M., Weng, J., Yang, A., Lu, W., Zhang, Y., Hou, L., ... & Deng, R. H. (2018). CrowdBC: A blockchain-based decentralized framework for crowdsourcing. IEEE Transactions on Parallel and Distributed Systems, 30(6), 1251-1266.
- [3] Yang, M., Zhu, T., Liang, K., Zhou, W., & Deng, R. H. (2019). A blockchain-based location privacy-preserving crowdsensing system. Future Generation Computer Systems, 94, 408-418.
- [4] Zou, S., Xi, J., Wang, H., & Xu, G. (2019). CrowdBLPS: A blockchainbased location-privacy-preserving mobile crowdsensing system. IEEE Transactions on Industrial Informatics, 16(6), 4206-4218.
- [5] Kadadha, M., Otrok, H., Mizouni, R., Singh, S., & Ouali, A. (2020). SenseChain: A blockchain-based crowdsensing framework for multiple requesters and multiple workers. Future Generation Computer Systems, 105, 650-664.
- [6] An, B., Xiao, M., Liu, A., Xu, Y., Zhang, X., & Li, Q. (2021). Secure Crowdsensed Data Trading Based on Blockchain. IEEE Transactions on Mobile Computing.

Publications

